

Lean Six Sigma Green Belt Certification, Course Notes

The Manufacturing Academy

What is Six Sigma?

- **Definition:** Sigma represents dispersion in a population.
- **Bell Curve:** Normal distribution used in statistics.
- **Height Example:** Illustrates the concept with adult males' heights.

Six Sigma Methodology

- **Data-Guided Approach:** Focuses on problem-solving.
- **Goal:** Achieve +/- 6 sigmas of a process output.
- **Population Percentage:** Varies with sigma levels.

Six Sigma and DMAIC

- **Continuous Improvement Model:** Centered around DMAIC.
- **Steps:** Define, Measure, Analyze, Improve, Control.

Six Sigma Tools

- **Tools:** SIPOC Diagram, Value Stream Map, Pareto Analysis, etc.
- **Goal:** Reduce process variation in projects.

Lean Manufacturing

- **Waste Reduction:** Focuses on doing more with less.
- **Customer Value:** Process steps add or fail to add value.

Lean Manufacturing Wastes

- **7 Mudas:** Transportation, Inventory, Motion, etc.
- **Benefits:** Lead time reduction, improved earnings, etc.

Lean and Six Sigma Similarities

- **Improvement Methodologies:** Both employ DMAIC.
- **Applications:** Can be used in various processes.
- **Complementarity:** Tools of both methodologies work well together.

Lean Manufacturing and 7 Mudras

- **Wastes:** Waiting, Overproduction, Overprocessing, Defects, Motion, Inventory, Transportation.
- **Impact:** Lead time extension, diminished earnings, etc.

Six Sigma and the 6 M's

- **Sources of Variation:** Machine, Method, Man, Material, Measurement, Mother Nature.
- **Objective:** Reduce variation in processes.

Cause and Effect Diagram

- **Tool:** Brainstorming tool for uncovering root causes.
- **6M's Diagram:** Utilizes Machine, Method, Material, etc.
- **Usage:** Helps categorize causes and analyze issues holistically.

Belt Colors and Levels of Mastery

- Belt colors in Six Sigma borrowed from martial arts
- Six Sigma has five belt colors: White, Yellow, Green, Black, and Master Black
- Lean Manufacturing and Lean Six Sigma follow the same belt system

White Belt

- Overview training
- Designed for stakeholders not involved in improvement projects
- Explains lean manufacturing and Six Sigma concepts

Yellow Belt

- Basic level of training
- Includes major concepts, problem-solving approaches, and statistical methods
- Provides support to improvement projects

Green Belt

- Intermediate level of training
- In-depth exposure to statistical and graphical tools
- Assist Black Belts in leading complex projects

Black Belt

- Advanced level of training
- Deep understanding of Six Sigma and Lean Manufacturing tools
- Lead complex projects and train Green and Yellow Belts

Master Black Belt

- Advanced leadership and technical acumen
- Drive organizational change and innovation
- Serve alongside top executives in setting organizational strategy

Calculating DPPM

- DPPM stands for Defective Parts Per Million
- DPPM calculation involves multiplying the decimal by 1,000,000
- Different percentages correspond to different DPPM values

Project Management Basics

- A project is a temporary endeavor to create a unique product, service, or result
- Projects have constraints: Scope, Budget, and Schedule
- Project phases include starting, organizing, conducting work, and closing the project

Business Finance Basics

- Lean Six Sigma projects tie into an organization's goals and strategy
- Financial success measured through methods like Payback Period and Return on Investment Analysis
- Inflation affects purchasing power and project value needs to outpace inflation

Team Building

- Lean Six Sigma is a team sport
- Team roles include Team Leader, Project Sponsor, Subject Matter Expert, Team Members, Finance Partner, and Champion
- Teams pass through stages of development: Forming, Storming, Norming, Performing

Selecting a Problem to Tackle

- Various sources provide suggestions for problems to tackle
- Pareto Analysis prioritizes problems based on the Pareto Principle
- Pareto Principle separates the "Significant Few" from the "Trivial Many"
- Example from quality reject data
- Overwhelmed by 39 problems
- Sorting by "Total Rejects"
- 4,335 out of 6,979 total rejects
- Sorting by "% Rejected of Total Sorted"
- Adding Cost per piece
- Sorting by "Total Cost of Rejects"
- Examples of parts with Total Sorted, Total Rejects, Cost per piece, and Total Cost of Rejects provided
- Pareto Analysis for problem prioritization
- Focus on significant issues for greater impact

Theory of Constraints

- Theory of Constraints
- Identifying the most limiting factor in a system
- Flow constraint in a system
- Importance of the flow through the narrowest constraint
- Iteration of Theory of Constraints
- Flow of the entire system based on the narrowest constraint
- Effects of system running at 80 Parts/Hour
- Starvation and accumulation of inventory in processes
- Improvements at constrained processes improve the system

Six Sigma Overview

- Six Sigma Tools overview
- Mix of art and science
- Various tools used in different areas
- Control charts, DMAIC approach
- Defining the problem in problem-solving
- Objectives of the Define phase
- Tools used in the Define phase like SIPOC Diagram, Cost of Quality, etc.
- Importance of defining the problem before resolving it

Value Stream Mapping

- Value Stream Mapping (VSM) in lean manufacturing
- Differentiating Value-Added and Non-Value-Added activities
- Essentiality of Value-Added activities
- Categorizing activities in Value Stream Mapping
- Essential and non-essential activities
- Reduce, eliminate, or retain Non-Value-Added activities
- Transmission Assembly at Avanti Motors Value Stream Map
- Involvement of Can Fastener XYZ Fluids, ACME Supply, JB Housings
- Detailed steps in Transmission Assembly at Avanti Motors
- Inclusion of Raw Storage, Sub-Assembly, Assembly, Test and Inspection, Packaging, and Storage steps

- Value Stream Mapping, Pt 2
 - Transmission Assembly at Avanti Motors
 - Components: Fastener XYZ, Fluids, Production Control
 - Process: Raw Storage, Sub-Assembly, Assembly, Test and Inspection, Packaging, Storage
 - Frequency: Monthly and Weekly releases

The Gemba

- The Gemba
 - Definition: "actual place" where work is done
 - Importance: Well-defined problems lead to effective solutions
 - Examples: Restaurant Kitchen, Hospital Operating Room, Car manufacturer Assembly Line
- The Gemba Walk
 - Benefits: Engage with employees, Boost morale, Foster collaboration
 - Leadership Role: Engage in Gemba Walks to drive positive change

Cost of Quality

- Cost of Quality
 - Definition: Extends beyond scrap costs
 - Model by Armand Feigenbaum: 4 Cost Buckets - Preventive, Appraisal, Internal, External
 - Examples: Employee Training, Quality Auditing, Warranty Claims

SIPOC

- SIPOC Diagram
 - Purpose: Identify elements of a business process
 - Acronym: Suppliers, Inputs, Process, Outputs, Customers
 - Usage: Define phase of improvement projects

RTY

- Rolled Throughput Yield (RTY)
 - Definition: Probability of defect-free product in a multi-step process
 - Calculation: $\text{Input to A1} * (1 - \text{A1 Defect Rate}) = \text{Input to A2}$

OEE

- Overall Equipment Effectiveness (OEE)
 - Definition: Measure of manufacturing productivity
 - Components: Availability, Performance, Quality
 - Calculation: $OEE = \text{Availability} \times \text{Performance} \times \text{Quality}$

TEEP

- Total Effective Equipment Performance (TEEP)
 - Definition: Measure of operational productivity
 - Components: Utilization, Availability, Performance, Quality
 - Calculation: $TEEP = \text{Utilization} \times \text{Availability} \times \text{Performance} \times \text{Quality} = \text{Utilization} \times OEE$

What are Descriptive Statistics?

- Descriptive Statistics
 - Purpose: Answer "What happened?"
 - Tools: Central tendency (mean, median, mode), Dispersion (range, standard deviation)

Breakeven Analysis

- Breakeven Analysis
 - Purpose: Estimate units for payoff, Justify initial investment based on cost savings
- Commercial painter considering purchasing a large sprayer for painting fences.
 - Sprayer costs \$6,000.
 - General contractors pay \$4 per foot for fence painting.
 - Cost for labor and material with sprayer is \$1.25 per foot.
- Need to calculate the number of feet of fence to paint to pay off the sprayer.
- Revenue calculation at \$4 per foot.
- Selling Price = $\$4 \times (\text{Length of Fence})$.
- Selling Price for Fence Painting graph provided.
- Total costs to paint fences calculation.
- Cost of Fence Painting = $\$6,000 + \$1.25 \times (\text{Length of Fence})$.
- Breakeven Point is where selling price equals costs.
- Selling Price and Cost of Fence Painting graph provided.

- Equating Cost of Fence Painting and Selling Price.
- Calculation for Breakeven Point.
- Breakeven Point = 2,182 Feet of Fence.
- Estimating capital investment justified by savings per unit.
- Example with inspection equipment saving \$2.75 per unit.
- Justified initial investment calculation: $\$2,500 \text{ units} * \$2.75/\text{unit} = \$6,875$.

Define the Problem

- Tools used in the Define phase.
- Importance of defining the problem thoroughly.
- Quote by Albert Einstein on problem-solving.
- Reiteration of tools used in the Define phase.
- Importance of defining the problem before resolving it.
- Examples of problem definition statements.
- Tips for defining a problem effectively.
- Value Stream Map for Transmission Assembly at Avanti Motors.
- Detailed layout of processes and flow in the assembly.
- Introduction of the problem-solving team members.
- Roles and expertise of Larry, Sandeep, Debra, and Juanita.

Measure the Problem

- Importance of measuring a problem for progress comparison.
- Objectives of the Measure phase.
- Tools used in the Measure phase for data collection.

SWOT Analysis

- Strategic planning tool
- Strengths, Weaknesses, Opportunities, Threats.
- Best when developed with cross-functional team over time.

Process Capability Analysis

- Definition and importance of Process Capability Analysis.
- Explanation of capability indices like Pp, Ppk, Cp, and Cpk.
- Calculation of Process Capability.

- **Specification Range**
 - Comes from engineering drawing or specifications
 - Calculated as $USL - LSL$
 - Example: Spec Range = $0.018 - 0.002 = 0.016$
- **Process Range**
 - Estimated by statistical analysis of measured values
- **Histogram Plotting**
 - First step in capability analysis
 - Helps visualize data distribution
- **Normal Distribution**
 - Data should be approximately normally distributed
 - Use visual comparison or statistical tests like Shapiro-Wilk
- **Parameters Calculation**
 - Sample mean and sample standard deviation
 - Mean estimates central tendency, SD estimates dispersion
- **Excel Calculation**
 - Sample mean: average formula
 - Sample standard deviation: stdev.s formula
 - Example: Sample mean = 37.01002, Sample SD = 0.0005772

- **Interpretation**
 - Specification Range vs. Process Range
 - Calculation: Process Range = $6 * SSD$
 - Example: Process Range = 0.0034632
- **Capability Indices**
 - Pp and Ppk
 - Pp compares spec range and process range
 - Ppk accounts for centeredness within spec range
- **Pp and Ppk Importance**
 - Quantitatively describe process capability

CTQ Characteristics

- **Critical to Quality (CTQ)**
 - Required for meeting product's functional requirements
 - Address Voice of the Customer (VOC)
- **CTQ Process Characteristics**
 - Ensure control over CTQ product characteristics
- **Examples**
 - Burst resistance, dimensional stability, color uniformity
 - Fill pressure, shot size, colorant concentration

Interviews and Observations

- **Qualitative Measurements**
 - Complement quantitative measurements
- **Interviews**
 - Open-ended questions, follow-ups
 - Avoid biased questions and inserting opinions
- **Observations**
 - Look for "Pain Points"
- **Empathizing**
 - Helps identify user needs and potential solutions

Stakeholders

- Definition of stakeholders in project management
- Stakeholders influence project success
- Bring knowledge and experience to the project
- Internal stakeholders: inside the organization
 - Project manager, project team, shop floor employees
- External stakeholders: outside the organization
 - End users, material suppliers, surrounding community
- Various stakeholders in a manufacturing project
 - Mfg. Manager, Maintenance, Quality, Engineering, Sales Manager, etc.
- Potential stakeholders in a project
 - President, Manufacturing Manager, Project Team, Equipment Suppliers, etc.
- Power-Interest Matrix for stakeholders
- Different levels of interest and power

- Strategies for managing stakeholders based on their power and interest levels

Scatterplots

- Scatterplot as a quality tool
- Shows the relationship between two variables
- Independent and dependent variables
- Example of a scatterplot with kPa and psi values
- Scatterplot example with independent and dependent variables
- Scatterplot example with air pressure and height of nail out of board
- Correlation in scatterplots
- Relation between statistical variables
- Correlation is not causation
- Excel-based video for building scatterplots in green belt class

Control Charts

- Run Charts and Control Charts
- Run Chart as a line graph over time
- Data example with sample numbers and values
- Control Charts with Control Limits
- Control Limits vs. Specification Limits
- Control Limits indicate expected process variation
- **Control Chart Constants:**
 - A2 is found on a table.
 - Formula for control chart constants involves the following parameters.
- **Sample Data:**
 - Provided sample data in a tabular format for 5 samples.
 - Each sample has 20 data points.
- **X-bar and R Values:**
 - X-bar values range from 0.6174 to 0.6193.
 - R values range from 0.0004 to 0.0021.
- Control chart constants can be found using a table.
- Sample data consists of 5 samples with 20 data points each.

- X-bar values range from 0.6174 to 0.6193, and R values range from 0.0004 to 0.0021.
- Control Charts and Control Limits
 - LCL = 0, UCL = 0.6
 - Control Limits formula: $\bar{X} \pm (\bar{R} * A2)$
- Sample data provided for 20 samples
 - X-bar values ranging from 0.6174 to 0.6195
 - R values ranging from 0.0004 to 0.0021

MSA

- Measurement Systems Analysis (MSA)
 - T.O.V. = $T + e$
 - Goals of MSA:
 - Determine portion of T.O.V. that is e
 - Determine components of e
 - Assess gage suitability for intended application
- MSA tools include:
 - Tape Measure
 - Digital Calipers
 - Digital Indicator Stand
 - Coordinate Measurement Machine
- Measurement Error and Tolerance levels:
 - Tape Measure: ± 0.1
 - Digital Calipers: ± 0.01
 - Digital Indicator with Stand: ± 0.001
 - Coordinate Measuring Machine (CMM): ± 0.000001
- Measurement errors related to central tendency:
 - Bias, Linearity, Stability
- Measurement errors related to variability:
 - Repeatability, Reproducibility, Resolution
- Precision to tolerance ratio and R&R template in Excel

The Measure Phase

- Importance of measuring a problem for baseline comparison
- Objectives of the Measure phase

- Tools used in the Measure phase
- Objectives of the Measure phase reiterated
- Examples of tools used in the Measure phase
- Examples of Measurement Statements
- Tips for measuring a problem
- Importance of accurate measurement in problem-solving

The Analyze Phase

- Analyze the Problem
 - Objectives of the Analyze phase
 - Identify process drivers, root causes, and sources of process variation
 - Tools used in the Analyze phase
 - Value Stream Analysis, Regression Analysis, Hypothesis Testing, etc.
 - Importance of not knowing too much about the problem when suggesting solutions

Curiosity

- Curiosity
 - Curiosity as a key intellectual characteristic of problem solvers
 - Definition of curiosity and its role in pushing past mental biases
 - Importance of curiosity in preparing the brain for learning
- How to awaken curiosity
 - Adopting a beginner's mindset
 - Practicing humility and separating symptoms from root causes
 - Goal of learning something new and adjacent learning

Cause and Effect

- Cause and Effect Diagram (Ishikawa or Fishbone diagram)
 - A brainstorming tool used to identify causes of a problem
 - Can be used before or after the measurement phase

5 Why

- 5 Why technique
 - Developed by Sakichi Toyoda in the 1930s

- Brainstorming technique to identify root causes by asking "Why" questions
- Limitations and benefits of the 5 Why technique
- Example of 5 Why technique in action
 - Series of "Why" questions to identify the root cause of a flat tire
- 5 Why, 3 Layers
 - Layers of the 5 Why technique: Occurrence, Escapement, Systemic
 - Examples of questions at each layer to identify root causes

Value Stream Analysis

- Value Stream Analysis
 - Example of a Value Stream Map for Transmission Assembly at Avanti Motors
 - Components of the Value Stream Map and their roles in the production process

Takt Time

- Takt time
 - Definition of Takt time as the maximum time needed to meet customer demand
 - Calculation of Takt time using Total Available Production Time and Average Customer Demand
- Takt time and its implications
 - Effects of cycle times slower or faster than Takt time on production
 - Importance of aligning cycle time with Takt time for minimizing waste and ensuring uniform flow

Total Costs of Inventory

- Lean Six Sigma Green Belt class aims to connect improvement tools with business objectives.
 - Reduction of non-value-added time like finished good storage and wait times leads to cost reduction.
- Components of Total Costs of Inventory:
 - Fixed Costs: Machine setup, material handling, administrative costs.
 - Holding Costs: Labor, material, utilities, operational costs, costs of capital.

Linear Regression

- Example of linear regression with independent and dependent variables.
- Height of nail out of board represented in a graph for regression analysis.

Qualitative Analysis

- Importance of qualitative research in understanding perceptions and feelings.
- Advantages of qualitative research: captures natural environment, leads to new problem ideas.
- Disadvantages: recollections may be incomplete, interviewees can be biased.
- Need for verification of qualitative findings to avoid confirmation bias.

Analyze the Problem

- Objectives of the Analyze phase in problem-solving.
- Tools used in the Analyze phase: Value Stream Analysis, Regression Analysis, Financial Analysis, etc.
- Importance of being open-minded to unexpected conclusions in analysis.

Improve the Problem

- Objectives of the Improve phase in problem-solving.
- Tools used in the Improve phase: 5S, Poka Yoke, Kanban, etc.
- Importance of implementing permanent corrective actions for improvement.

Prototyping

- Importance of prototyping in problem-solving process.
- Prototype > Pilot > Production model for product design and process verification.
- Definition and significance of a prototype in testing and validating ideas.

Change Management

- Importance of mindset change in change management.
- Involvement of managers in change processes for effective implementation.

5S

- Definition and purpose of 5S in creating an organized workplace.
- Steps in the 5S process: Sort, Set in order, Shine, Standardize, Sustain.
- Benefits of 5S: improved efficiency, productivity, and employee morale.

- Arrange articles logically for optimal workflow
- Utilize shadow boards for organization
- Clean work and storage areas thoroughly
- Wash, wax, and repaint surfaces
- Replace damaged ceiling tiles and repair broken handles
- Establish procedures for tool and document retrieval
- Implement best practices to prevent damage
- Set up calibration and maintenance programs
- Aim for continuous improvement in efficiency and integrity
- Encourage commitment within the team

Maintenance

- Three modes: Corrective, Preventive, Predictive
- Examples of each mode and their importance

Cellular Manufacturing

- Benefits of Cellular Manufacturing (CM)
- Implementation with other Lean Manufacturing (LM) improvements
- Advantages of Cellular Layout over Traditional Layouts
- Features of Cellular Layouts like U-Shaped Flow Lines

Design for Assembly

- Definition and pioneers of Design for Assembly (DFA)
- Guidelines for DFA to improve assembly process

Poka Yoke

- Meaning of Poka Yoke and its purpose
- Application in consumer products and industrial equipment

Single Minute Exchange of Die (SMED)

- Goal of reducing machine change-over time
- Core concept of SMED and its relation to Toyota Production System

- **Single Minute Exchange of Die (SMED)**
 - Methodology to reduce machine change-over time to under 10 minutes.
 - Core component of the Toyota Production System.
 - Critical concept: Move activities to when the machine is running.
- **Economic Order Quantity**
 - Involves holding costs, fixed costs, total cost, annual costs, and quantity per production run.
- **SMED Projects**
 - Steps: Select a setup process, record the process, assign "elements" to steps.
- **SMED Elements**
 - Classify elements as "External" (while machine is running) or "Internal" (machine stopped).
- **SMED Steps Continued**
 - Re-order steps to move External elements outside machine downtime.
 - Example elements and their classification provided.
- **SMED Improvement Techniques**
 - Engineer Internal elements into External elements.
 - Techniques: Advance Preparation, Make adjustments prior to changeover, Convert to standard set-ups.
- **Quick Die Change Hardware**
 - Mention of 1200# DIE.

Process Improvement Tools

- **Problem Improvement Tools**
 - Examples: 5S, Cellular manufacturing, Poka Yoke, SMED, etc.
 - Quote by Mark Twain on continuous improvement.
- **Process Improvement**
 - Tips on improving a problem iteratively.
 - Example of improving a traditional process layout.

Control Phase

- **Control Phase Objectives**
 - Lock in solutions and prevent recurrence.
 - Tools used: Andon, Training, Control Charts, etc.

Standardized Work

- **Standardized Work / Training**
 - Importance of written standards in training.
 - Training standard as a "living document."

LPA

- **Layered Process Auditing (LPA)**
 - Management tool to verify work against standards.
 - Importance and benefits of LPAs.
- **LPA Structure**
 - Process audits at job site by various layers of an organization.
 - Frequency and roles defined.
- **LPA Benefits**
 - Reduction of waste and safety incidents.
 - Reinforcement of key process steps and interaction improvement.

Process Control Plan

- **Process Control Plan**
 - Document describing quality control methods for each process step.
 - Includes control methods, responsible parties, and inspection frequencies.
- **Process Control Plan Characteristics**
 - Detailed breakdown of process control plan elements.
- **Process Control Plan Documents**
 - Coupling with Process Flow Diagram and Process Failure Modes and Effects Analysis.

Contingency Planning

- Contingency: a provision for an event that's timing cannot be predicted
- Contingency planning is the process of creating a backup plan to address risks to a project, business, or organization.
- Contingency planning is critical in manufacturing.

- A robust contingency plan considers internal and external risks from a variety of sources.
- Process improvement is centered around the customer's experience.

Andon

- **Andon System**
 - Definition and components of the Andon system.

Checklists

- **Checklists**
 - Simple tools for reminding users of tasks.
 - Widely used in manufacturing.

Visual Aids

- **Visual Aid / Quality Alert**
 - Coupled with training and part of permanent work instructions.
 - Use of Quality Alert form for distributing quality-related information.

Control Charts

- Control Charts
 - Control Limits: $LCL = 0$, $UCL = 0.6$
 - Formula for Control Limits: $X \pm (R * A2)$
- Sample data provided for 5 groups with 20 samples each
- X-bar values and R values calculated for each group
- Importance of verifying control limits periodically
- Recalculation of control limits after process improvements

Control Phase

- Control Phase example after successful corrective actions:
 - Updated work instruction
 - Training operators on new instruction
 - Addition of an andon light stack for alerts
- Control phase actions similar to corrective actions
- Control phase often done alongside the improvement phase
- Importance of explaining